

PHY201: Waves & Optics

Problem Set 3

Aug 30, 2025

1. $m = 0.20 \text{ kg}$, $k = 80 \text{ N m}^{-1}$, $b = 4 \text{ N s m}^{-1}$. Define the natural frequency and damping rate

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{80}{0.2}} = 20 \text{ s}^{-1}, \quad \gamma = \frac{b}{m} = \frac{4}{0.2} = 20 \text{ s}^{-1}.$$

Since $\beta < \omega_0$, the system is underdamped with

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2/4} = \sqrt{400 - 100} = \sqrt{300} = 10\sqrt{3} \approx 17.3205 \text{ s}^{-1}.$$

(a)

$$A(\Omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}}, \quad \tan \delta(\omega) = \frac{\gamma\omega}{\omega_0^2 - \omega^2}.$$

$F_0 = 2 \text{ N}$ and $\omega = 30 \text{ s}^{-1}$. Hence, calculate $A \approx 1.28 \text{ cm}$ and $\delta \approx 2.26 \text{ rad}$.

2. *Given:* (1) A horizontal push $= mg$ produces a static compression $h \Rightarrow kh = mg \Rightarrow k = \frac{mg}{h}$.
(2) The viscous resistive force equals mg at speed $u \Rightarrow bu = mg \Rightarrow b = \frac{mg}{u}$.

(a) With displacement $x(t)$ from equilibrium:

$$m\ddot{x} + b\dot{x} + kx = 0 \quad \Rightarrow \quad m\ddot{x} + \frac{mg}{u}\dot{x} + \frac{mg}{h}x = 0.$$

Define $\omega_0^2 = \frac{k}{m} = \frac{g}{h}$, $\gamma = \frac{b}{m} = \frac{g}{u}$.

Assume $u = 3\sqrt{gh}$.

(b) Damped frequency.

$$\omega = \sqrt{\omega_0^2 - \frac{\gamma^2}{4}} = \sqrt{\frac{g}{h} - \frac{g^2}{4u^2}} = \frac{\sqrt{35}}{6} \sqrt{\frac{g}{h}}.$$

(c) Energy $1/e$ time. Energy decays as $\bar{E} \sim e^{-\Gamma t}$, so $t_E = 1/\Gamma = u/g = 3\sqrt{h/g}$:

$$t_E = 3\sqrt{\frac{h}{g}}.$$

(d) Quality factor.

$$Q = \frac{\omega_0}{\Gamma} = \frac{\sqrt{g/h}}{g/u} = \frac{u}{\sqrt{gh}} = 3.$$

(e) Impulsive start; phase δ . With $x(0) = 0$ and a positive initial velocity from an impulse, the underdamped response $x = Ae^{-\gamma t/2} \cos(\omega t - \delta)$ requires $A \cos \delta = 0$ and $\dot{x}(0) > 0$, hence

$$\delta = \frac{\pi}{2}, \quad x(t) = Ae^{-\gamma t/2} \sin(\omega t).$$

(f) Driven amplitude at $\omega = \sqrt{2g/h}$ for $F(t) = mg \cos \omega t$:

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\gamma\omega)^2}} = \frac{g}{\sqrt{\left(\frac{g}{h} - \omega^2\right)^2 + \left(\frac{g}{u}\omega\right)^2}}.$$

With $\omega_0^2 = g/h$, $\omega^2 = 2g/h$, $u = 3\sqrt{gh}$,

$$\omega_0^2 - \omega^2 = -\frac{g}{h}, \quad \Gamma\omega = \frac{g}{u} \sqrt{\frac{2g}{h}} = \frac{\sqrt{2}}{3} \frac{g}{h},$$

$$A = \frac{3h}{\sqrt{11}}.$$

3. For steady-state motion $x(t) = X \cos(\omega t - \delta)$, $v(t) = -X\omega \sin(\omega t - \delta)$:

(a) Instantaneous kinetic energy:

$$T(t) = \frac{1}{2}mv^2 = \frac{1}{2}mX^2\omega^2 \sin^2(\omega t - \delta).$$

(b) Instantaneous potential energy (linear spring):

$$U(t) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega_0^2 X^2 \cos^2(\omega t - \delta).$$

(c) Ratio of cycle-averaged energies:

$$\langle T \rangle = \frac{1}{4}mX^2\omega^2, \quad \langle U \rangle = \frac{1}{4}mX^2\omega_0^2 \Rightarrow \frac{\langle T \rangle}{\langle U \rangle} = \left(\frac{\omega}{\omega_0}\right)^2.$$

(d) $\langle T \rangle = \langle U \rangle$ when $\omega = \omega_0$; then

$$E(t) = T + U = \frac{1}{2}mX^2\omega_0^2 \quad (\text{constant in time}).$$

(e) Time variation of total energy; when constant.

$$E(t) = \frac{1}{2}mX^2[\omega^2 \sin^2 \theta + \omega_0^2 \cos^2 \theta] = \frac{1}{4}mX^2(\omega^2 + \omega_0^2) + \frac{1}{4}mX^2(\omega_0^2 - \omega^2) \cos(2\omega t - 2\delta).$$

Thus E oscillates at 2ω about $\langle E \rangle = \frac{1}{4}mX^2(\omega^2 + \omega_0^2)$ and is constant in time exactly when $\omega = \omega_0$.